Sri SAI RAM ENGINEERING COLLEGE
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Sairam INSTITUTIONS

SAIRAM
DIGITAL RESOURCES

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YEAR II SEM III

LIVE-IN-LAB PROJECT WORK

Project Title : FULLY AUTOMATED ROBOTIC SYSTEMS

GUIDE NAME : Dr. P . Rayavel

3 GOOD HEALTH AND WELL-BEING

FULLY AUTOMATED ROBOTIC SYSTEMS

DOMAIN : MACHINE LEARNING AND ROBOTICS

Primary Goal No : 09 – Industry, Innovation and Infrastructure

Secondary Goal No : 08 – Decent work and economic growth

Tertiary Goal No : 17 – Partnership for all the goals

JUSTIFICATION FOR SDG & SAP No.

SDG 9: Industry, Innovation, and Infrastructure :

•**Objective**: Foster innovation and build resilient infrastructure.

•**Impact**: Developing a robot that integrates advanced technologies like AI and IoT contributes to innovation.

It supports the creation of smart classrooms and workplaces, promoting technological advancement and efficiency.

SDG 16: Peace, Justice, and Strong Institutions :

•**Objective**: Promote transparent and accountable institutions.

•**Impact**: An automated logging system enhances transparency and accountability in both educational and corporate institutions.

It ensures accurate record-keeping and reduces the potential for human error or manipulation.

SDG 8: Decent Work and Economic Growth

•**Objective**: Increase productivity and efficiency in the workplace.

•**Impact**: In a corporate setting, the robot can log employee attendance, manage schedules, and track performance metrics.

CORE OBJECTIVES OF THE PROJECT

- To develop a fully automated robotic system(FARS) that integrates face recognition using machine learning, Real time data analytics, cloud based database system and IoT technologies to enhance campus security and operational efficiency.
- This system will automate the identification of individuals, manage access control, and provide real-time data to administrators.

PROBLEM STATEMENT

- The need for enhanced security and operational efficiency in management is critical.
- Traditional security systems often rely on manual processes, leading to human errors, delays in response times, and increased operational costs.
- the growing complexity of management environments, coupled with evolving security threats, necessitates innovative solutions that can proactively identify risks, automate routine tasks, and provide real-time insights for informed decision-making.

Abstract

- This system uses machine learning to automatically monitor and record key activities in schools and workplaces, offering real-time data and insights to improve decision-making and efficiency.

- It's designed to integrate easily with existing systems, prioritize data privacy, and adapt to changing needs, making it a valuable tool for optimizing resources and boosting productivity.

LITERATURE REVIEW

S.No.	Title of the Journal Article	Authors	Year of Publication	Problem Defined	Drawback
01	"Face Recognition Systems in Security Applications"	Mathew Turk and Alex Pentland	2018	The study highlights the limitations of traditional security systems and the need for advanced face recognition technologies.	The existing systems often struggle with accuracy in diverse lighting conditions and require significant computational resources.
02	"Machine Learning for IoT Security"	Jane Smith	2020	This paper discusses the challenges of securing IoT devices and networks..	The integration of machine learning in IoT security is still in its infancy, with issues related to data privacy and real-time processing..

S.No.	Title of the Journal Article	Authors	Year of Publication	Problem Defined	Drawback
03	"Wearable Computing and Augmented Reality for Enhanced Accessibility"	Davis.L., & Kim.J.	2019	This study explores the potential of wearable computing and AR in improving accessibility for individuals with cognitive disabilities, focusing on real-time feedback and contextual information.	The article points out that while wearable AR devices offer potential, issues such as user comfort and device accuracy remain challenging.
04	"Real-Time Object Recognition and Description Systems for the Visually Impaired"	Lee.C., & Martinez.A	2022	The research investigates systems that use real-time object recognition to aid visually impaired users, providing descriptions of objects within their environment.	A significant drawback identified is the latency in object recognition and description, which affects the real-time usefulness of such systems.

EXISTING SYSTEM DESIGN

- **Manual Monitoring and Surveillance**
  - **Security Personnel**: Guards patrol and monitor feeds, prone to human error.
  - **CCTV Cameras**: Require continuous supervision, lack intelligent detection.
- **Access Control Systems**
  - **ID Cards and Badges**: Prone to loss, theft, and duplication.
  - **Keypad Entry Systems**: Vulnerable to shared or observed PIN codes.
- **Alarm Systems**
  - **Intrusion Alarms**: Prone to false alarms, triggering on motion detection.
  - **Fire Alarms**: Focused on fire safety, not comprehensive security.
- **Data Storage and Retrieval**
  - **Local Servers**: Difficult real-time access and analysis.
  - **Manual Review**: Time-consuming, impractical for immediate threat detection.
- **Communication Systems**
  - **Intercoms and Hotlines**: Outdated, limited efficiency in emergencies.
  - **Lack of Real-Time Alerts**: Reduces effectiveness of immediate responses.

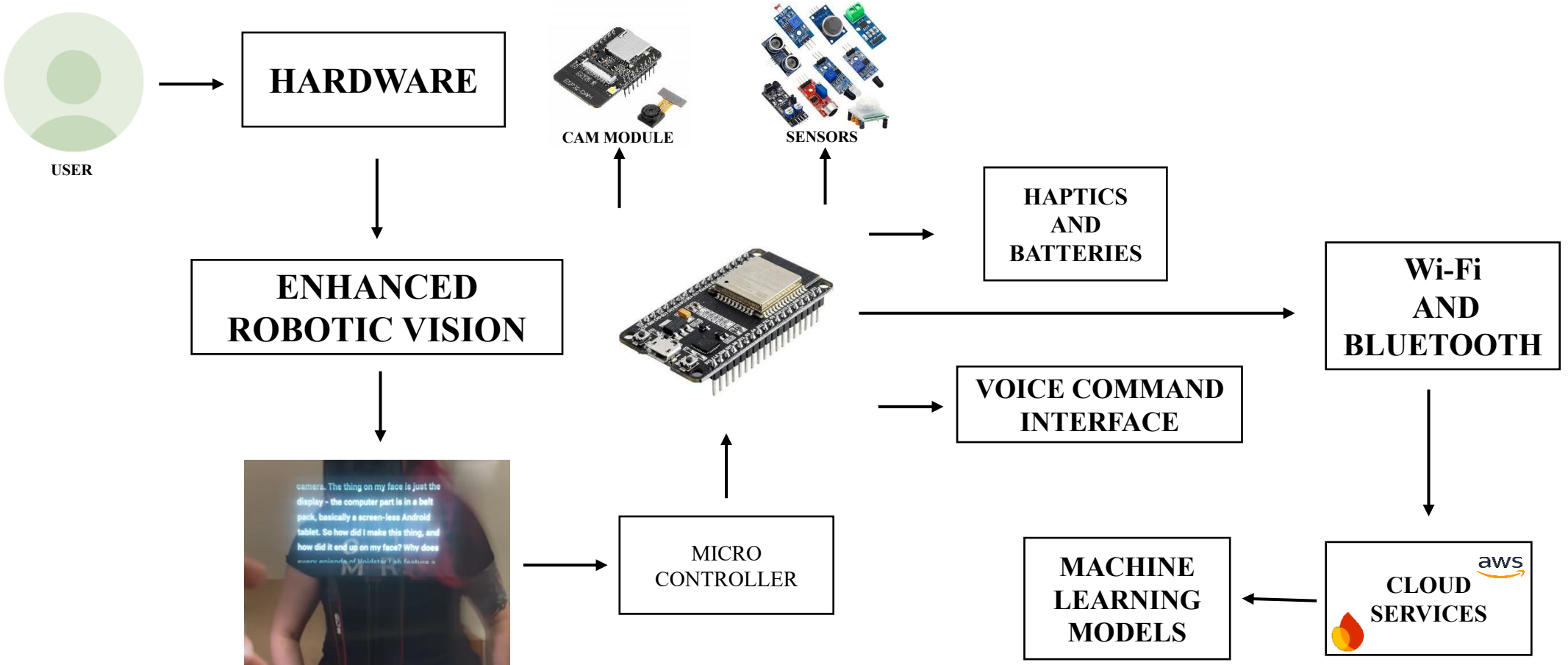
DESIGN OF THE PROPOSED SOLUTION

The proposed FARS system includes the following components:

- **Hardware**: NodeMCU ESP32, IR sensors, camera modules, networking equipment (routers, hubs), and Raspberry Pi.
- **Software**: Python for OpenCV face recognition, Firebase Cloud Server for authentication and data storage, HTML/CSS/JavaScript for the web/admin interface.
- **Libraries/Frameworks**: OpenCV, TensorFlow, Firebase SDK.

The system will use machine learning algorithms to improve face recognition accuracy and adapt to changing conditions. It will integrate with existing campus networks for seamless communication and coordination.

SYSTEM ARCHITECTURE



BUSINESS MODEL CANVAS

Key Partners

NGOs: Organizations working with differently-abled individuals.

Research Institutes: Institutions focusing on assistive technologies and AR research.

Educational Institutions: Schools, colleges, and universities.

Key Activities

Product development, R&D, marketing, customer support.

Key Resources

Financial capital, technical expertise, manufacturing facilities.

Value Proposition

Enhanced Independence: Provides differently-abled individuals with greater autonomy in navigation and object recognition.

Real-Time Assistance: Instant haptic feedback and guidance through AR overlays and voice commands.

Cost-Effective Solution: Affordable pricing compared to traditional assistive technologies.

Customer Relationship

Personal Assistance: Dedicated support for setup and troubleshooting.
Feedback Systems: Regular surveys and feedback mechanisms to gather user input for improvements.

Channels

Direct sales, online platforms, healthcare providers, educational institutions.

Customer Segments

Visually Impaired Individuals: Primary target users who will benefit from navigation and object recognition features.

NGOs: Organizations working with differently-abled communities.

Government Agencies: Programs focused on enhancing accessibility and independence for differently-abled individuals.

Educational Institutions: Schools and universities looking to provide inclusive education tools.

Cost Structure

Production cost roughly estimated : $199.30 = ₹16,698.52

Revenue Streams

Product sales, subscription services, partnerships, grants, customization services.

PROPOSED BUDJET

S.NO	NAME	AMOUNT
01.	Hardware Components (ESP BOARDS, Micro - Controllers) :	$35.81
02.	Manufacturing :	$14.33
03.	Assembly :	$9.55
04.	Software development :	$23.88
05.	Database management (AWS, AZURE) :	$12.65 (per month)
06.	Generation of ML model, AR Integration :	$41.94
07.	Maintenance :	$11.94
08.	Customer Support :	$14.33
09.	Initial Marketing :	$17.31
10.	Miscellaneous Components (sensors, wiring):	$17.91
	GRAND TOTAL :	$199.35 = ₹16,698.52

FUTURE ENHANCEMENT

•Scalability :

•**Expansion to Larger Areas**: The system can be extended to cover more extensive areas, such as additional campus buildings, parking lots, and nearby off-campus locations, ensuring comprehensive security coverage.

•Advanced Analytics :

•**Predictive Analytics for Threat Identification**: Implementing machine learning models that analyze historical data to predict potential security threats, enabling proactive measures to prevent incidents.

•**Behavioral Analysis**: Using advanced analytics to study the behavior patterns of individuals, identifying anomalies that may indicate suspicious activities or security breaches.

•Mobile Integration :

•**Administrator Mobile App**: Developing a mobile application that allows administrators to monitor live feeds, receive real-time alerts, and control the security system remotely, enhancing their ability to respond quickly to incidents.

•Enhanced Privacy :

•**Advanced Encryption Methods**: Incorporating state-of-the-art encryption techniques to secure data transmission and storage, ensuring that sensitive information remains protected from unauthorized access.

•**Data Anonymization**: Implementing anonymization protocols to protect the privacy of individuals captured in surveillance footage, making it possible to use the data for analysis without compromising personal privacy.

REFERENCE

- **Title of Journal Article**: "Real-time Face Detection and Recognition Using Improved Viola-Jones Algorithm and Deep Learning"
  - **Author**: A. Khraisat, I. Alsmadi, A. Awajan
  - **Problem Defined**: The paper addresses the need for real-time and accurate face detection and recognition systems.
  - **IEEE Journal**: IEEE Xplore Link

- **Title of Journal Article**: "IoT-Based Smart Surveillance Security System Using Deep Learning for Home Automation"
  - **Author**: H. N. Dai, R. C. Y. Ngai, M. Lyu
  - **Problem Defined**: The study focuses on the implementation of a smart surveillance system using IoT and deep learning technologies.
  - **IEEE Journal**: IEEE Xplore Link

Sairam SDG SOLVEATHON 4.0

Sri SAIRAM ENGINEERING COLLEGE INSTITUTE OF TECHNOLOGY

Autonomous Institutions | Affiliated to Anna University & Approved by AICTE, New Delhi
Accredited by NBA and NAAC "A+" | An ISO 21001:2018 Certified and MHRD NIRF ranked institution
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Imagine the Future and
Make it happen!

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Together let's build a better world where there is NO POVERTY and ZERO HUNGER. We have GOOD HEALTH AND WELL BEING QUALITY EDUCATION and full GENDER EQUALITY everywhere. There is CLEAN WATER AND SANITATION for everyone. AFFORDABLE AND CLEAN ENERGY which will help to create DECENT WORK AND ECONOMIC GROWTH. Our prosperity shall be fuelled by investments in INDUSTRY, INNOVATION AND INFRASTRUCTURE that will help us to REDUCE INEQUALITIES by all means. We will live in SUSTAINABLE CITIES AND COMMUNITIES. RESPONSIBLE CONSUMPTION AND PRODUCTION will help in healing our planet. CLIMATE ACTION will reduce global warming and we will have abundant, flourishing LIFE BELOW WATER, rich and diverse LIFE ON LAND. We will enjoy PEACE AND JUSTICE through STRONG INSTITUTIONS and will build long term PARTNERSHIPS FOR THE GOALS.

For the goals to be reached, everyone needs to do their part: governments, the private sector, civil society and **People like you.**

Together we can...

Sai Prakash Leo Muthu
Chairman & CEO – Sairam Institutions

Sairam RAISE Inspiring PEOPLE | Sairam EOMS Improving PROCESS | Sairam SDG ACTION PROGRAM Instilling PURPOSE | Sairam Innovation Eco-system | SAIRAM TECHNO INCUBATOR FOUNDATION | INSTITUTION'S INNOVATION COUNCIL | StartupTN

"We must be the change that we wish to see in the world"

- Mahatma Gandhi

Thank you

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Sairam RAISE
Inspiring PEOPLE

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